

Artificial Intelligence

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Learning Outcomes of Lecture 05

- Understand the related concepts of constraint satisfaction problem
- Understand the motivation of studying constraint satisfaction problem
- Master the strategies to speed-up the solving of CSP
- Understand the related concepts of games
- Understand the motivation of studying games
- Master a typical problem-solving method for games
- Master the strategies to speed-up the solution search

Agenda

- Machine learning
- Clustering
- Classification

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Machine Learning

■ Definition

- ❑ A branch of artificial intelligence
- ❑ Computational model that can learn from experience in the environment with respect to improve the performance of some tasks

■ According to the feedback from environment

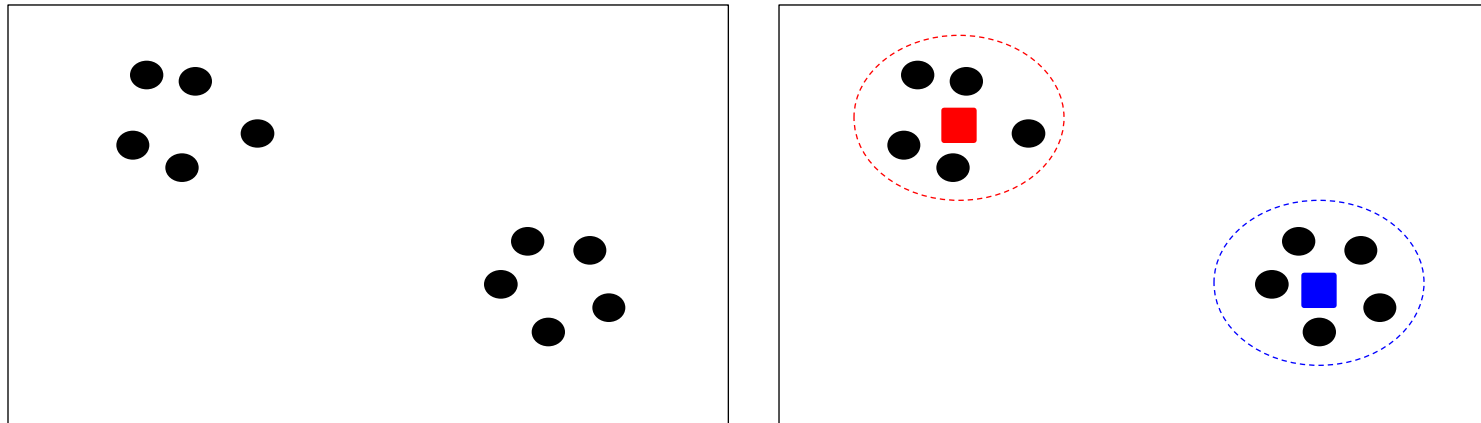
- ❑ Unsupervised learning
- ❑ Supervised learning
- ❑ Reinforcement learning

Agenda

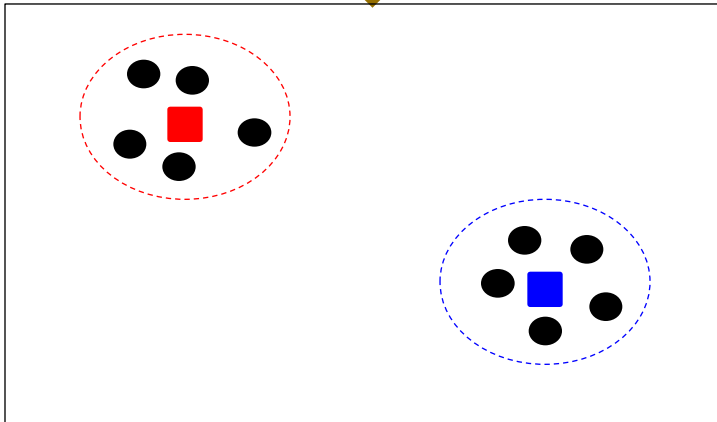
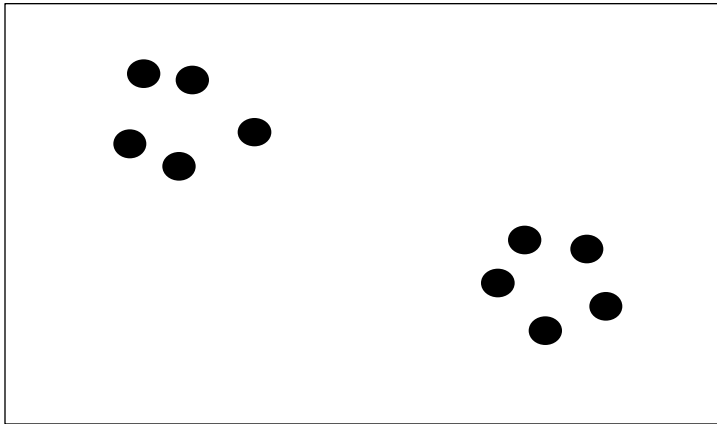
- Machine learning
- Clustering
- Classification

Clustering

- Clustering
 - Partition a given dataset into groups based on specified features so that the data points within a group are more similar to each other than the points in different groups
 - A typical unsupervised learning problem
 - Different clustering algorithms lead to different results



K-means clustering



Given

a set of observation $(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$,
where each observation is
a d -dimensional real vector

Aim to

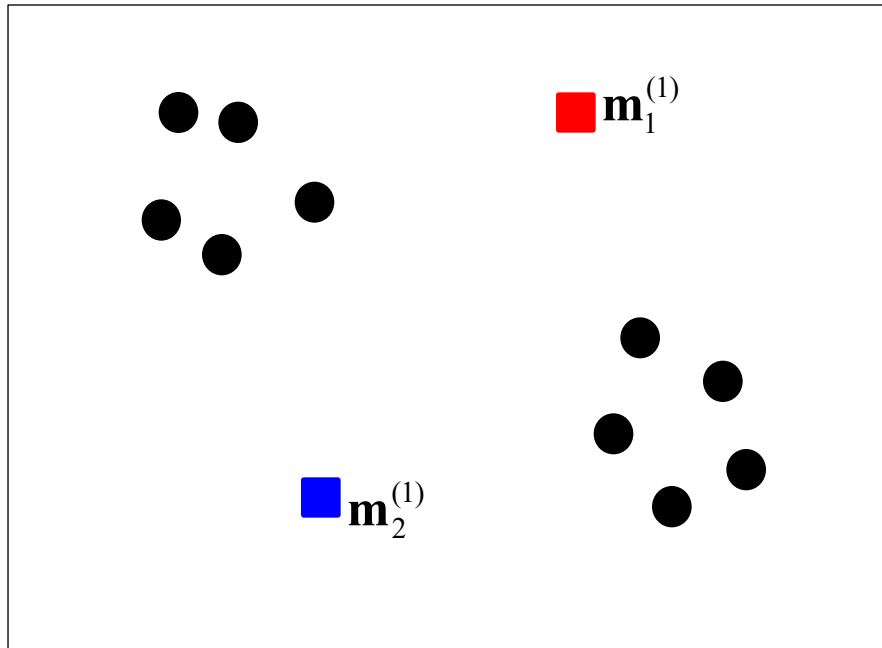
partition observations into K sets
($K < N$), $\mathbf{S} = \{S_1, \dots, S_K\}$
so as to minimize
the within-cluster sum of squares:

$$\arg \min_{\mathbf{S}} \sum_{i=1}^K \sum_{\mathbf{x}_j \in S_i} \|\mathbf{x}_j - \mu_i\|^2$$

where μ_i is the mean of S_i .

K means: details

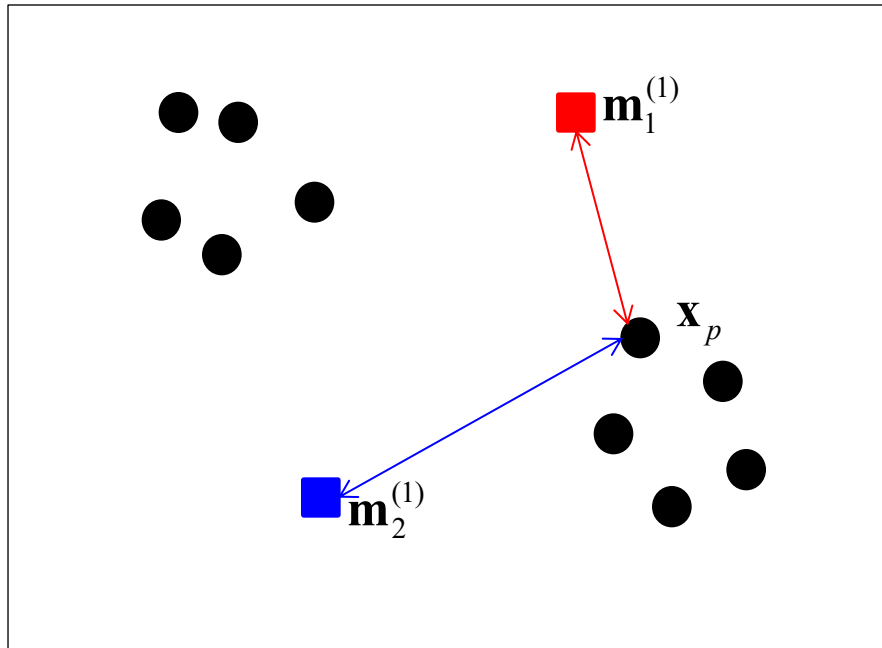
First iteration



K means begin with an initial guess to the centers: $\mathbf{m}_1^{(1)}$ and $\mathbf{m}_2^{(1)}$

K means: details

First iteration



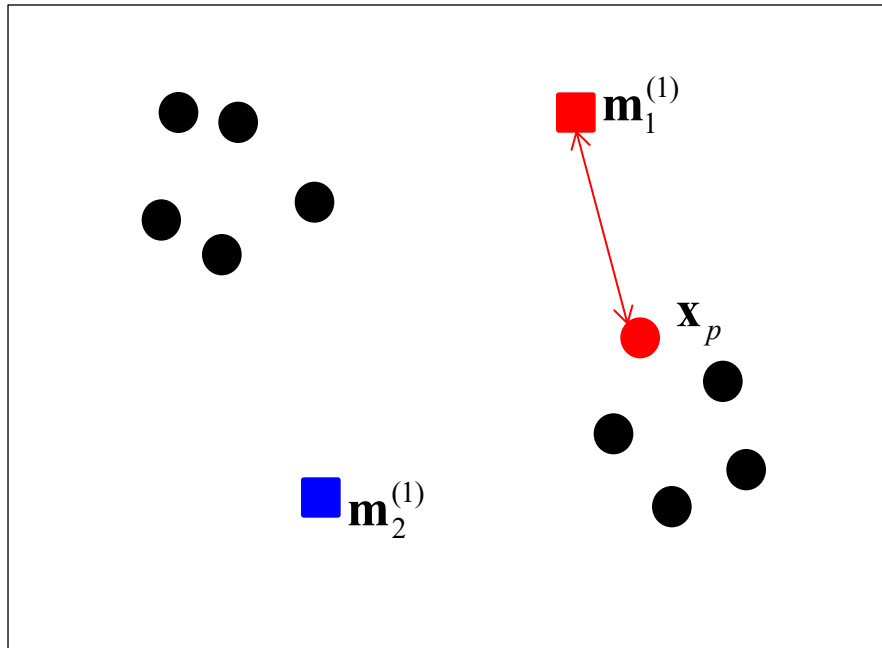
Calculate the Euclidean distance between each point and the centers

$$d^2(\mathbf{x}_p, \mathbf{m}_1^{(1)}) = \|\mathbf{x}_p - \mathbf{m}_1^{(1)}\|^2 = \sum_i (x_{pi} - m_{1i}^{(1)})^2$$

$$d^2(\mathbf{x}_p, \mathbf{m}_2^{(1)}) = \|\mathbf{x}_p - \mathbf{m}_2^{(1)}\|^2 = \sum_i (x_{pi} - m_{2i}^{(1)})^2$$

K means: details

First iteration



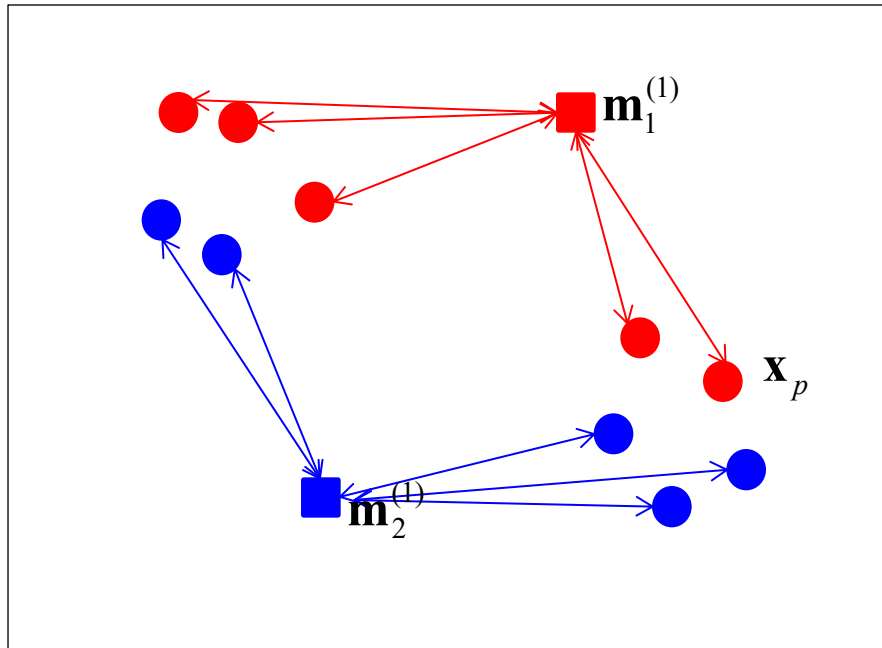
Assign the data point to the center with smallest distance:

$$d^2(\mathbf{x}_p, \mathbf{m}_1^{(1)}) < d^2(\mathbf{x}_p, \mathbf{m}_2^{(1)})$$

Then $\mathbf{x}_p$ is assigned to $\mathbf{m}_1^{(1)}$

K means: details

First iteration



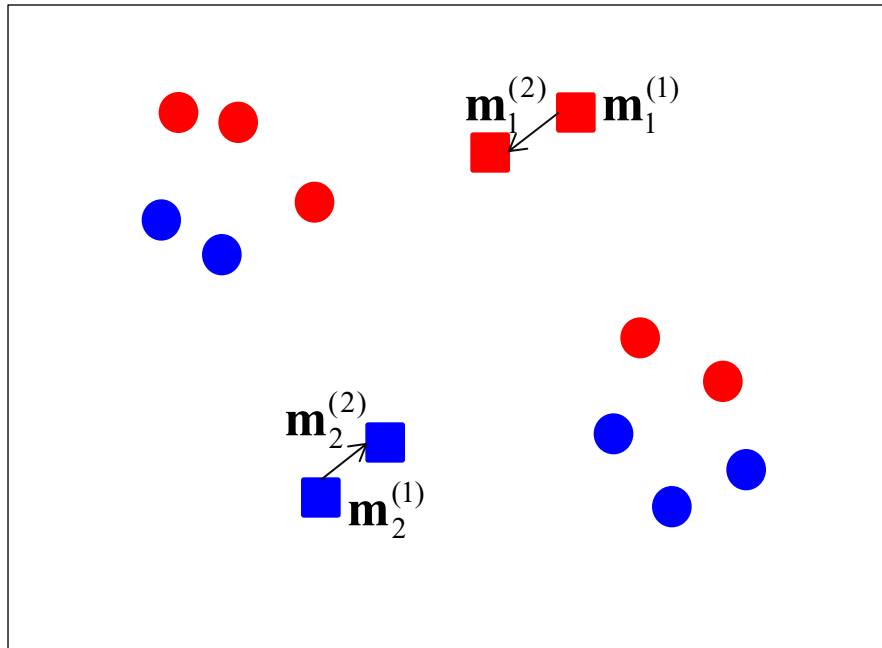
Assign all the other points to the centers in the same way as $\mathbf{x}_p$

$$S_1^{(1)} = \{\mathbf{x}_p : \|\mathbf{x}_p - \mathbf{m}_1^{(1)}\|^2 \leq \|\mathbf{x}_p - \mathbf{m}_2^{(1)}\|^2\}$$

$$S_2^{(1)} = \{\mathbf{x}_p : \|\mathbf{x}_p - \mathbf{m}_2^{(1)}\|^2 \leq \|\mathbf{x}_p - \mathbf{m}_1^{(1)}\|^2\}$$

K means: details

First iteration



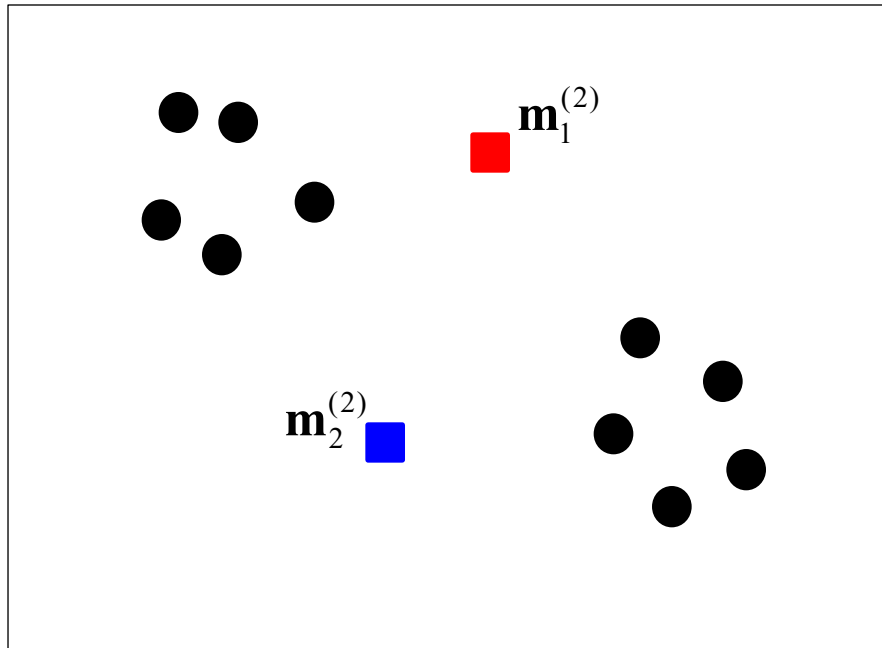
Update new centers by averaging the data points in each set:

$$\mathbf{m}_1^{(2)} = \sum_{\mathbf{x}_p \in S_1^{(1)}} \mathbf{x}_p / |S_1^{(1)}|$$

$$\mathbf{m}_2^{(2)} = \sum_{\mathbf{x}_p \in S_2^{(1)}} \mathbf{x}_p / |S_2^{(1)}|$$

K means: details

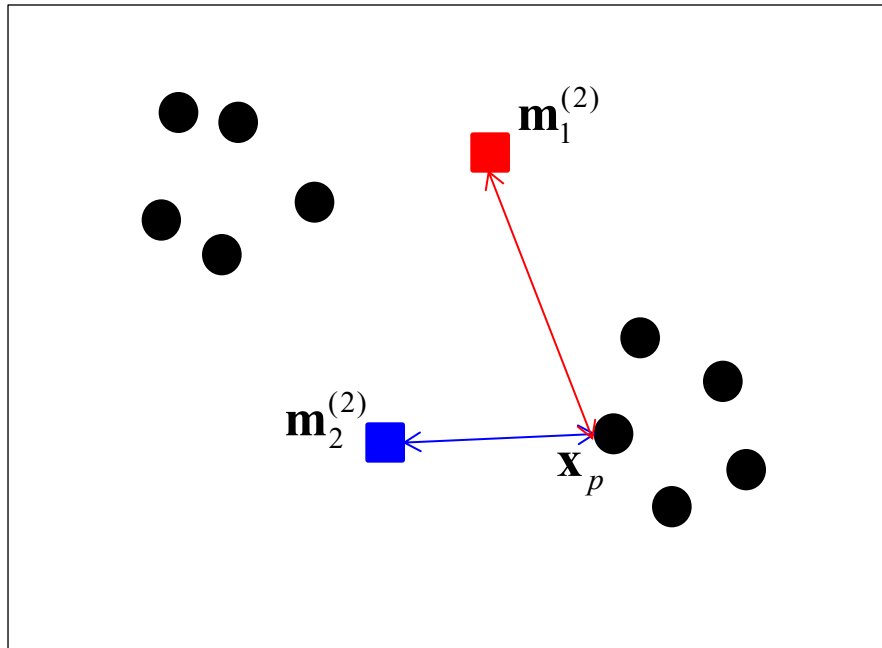
First iteration



The result after first iteration

K means: details

Second iteration



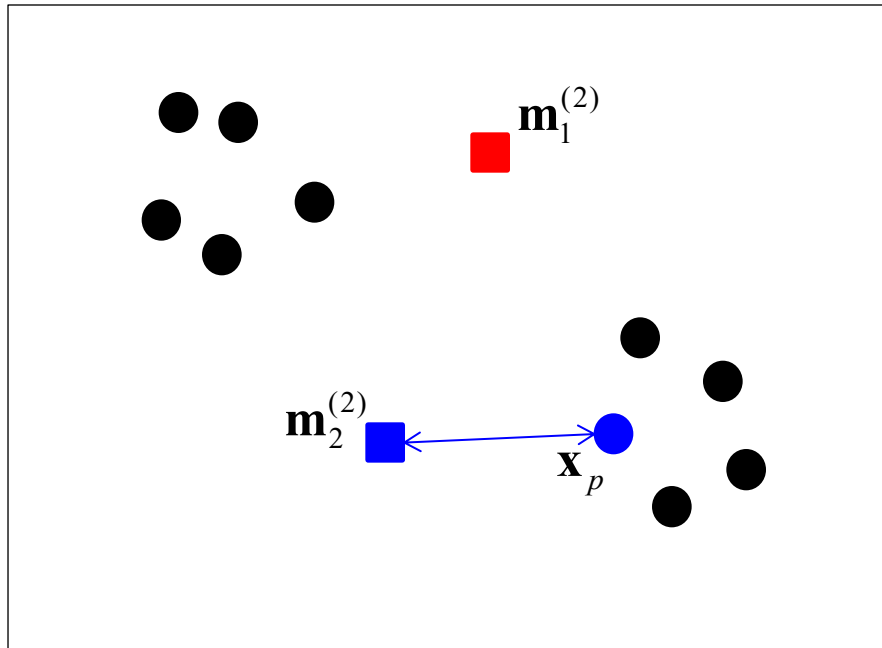
Calculate the Euclidean distance between each point and the centers

$$d^2(\mathbf{x}_p, \mathbf{m}_1^{(2)}) = \|\mathbf{x}_p - \mathbf{m}_1^{(2)}\|^2 = \sum_i (x_{pi} - m_{1i}^{(2)})^2$$

$$d^2(\mathbf{x}_p, \mathbf{m}_2^{(2)}) = \|\mathbf{x}_p - \mathbf{m}_2^{(2)}\|^2 = \sum_i (x_{pi} - m_{2i}^{(2)})^2$$

K means: details

Second iteration



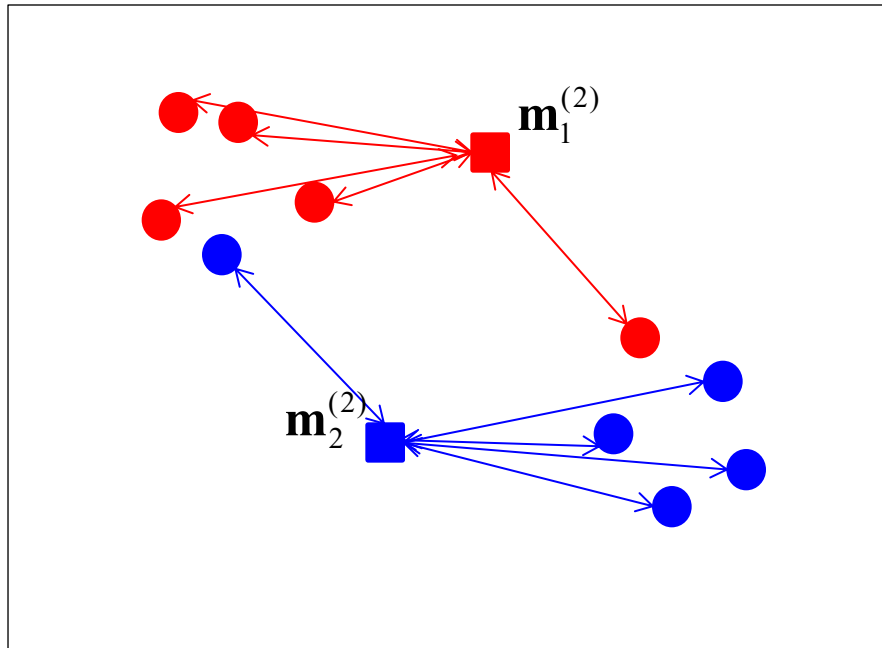
Assign the data point to the center with smallest distance:

$$d^2(\mathbf{x}_p, \mathbf{m}_2^{(2)}) < d^2(\mathbf{x}_p, \mathbf{m}_1^{(2)})$$

Then $\mathbf{x}_p$ is assigned to $\mathbf{m}_2^{(2)}$

K means: details

Second iteration



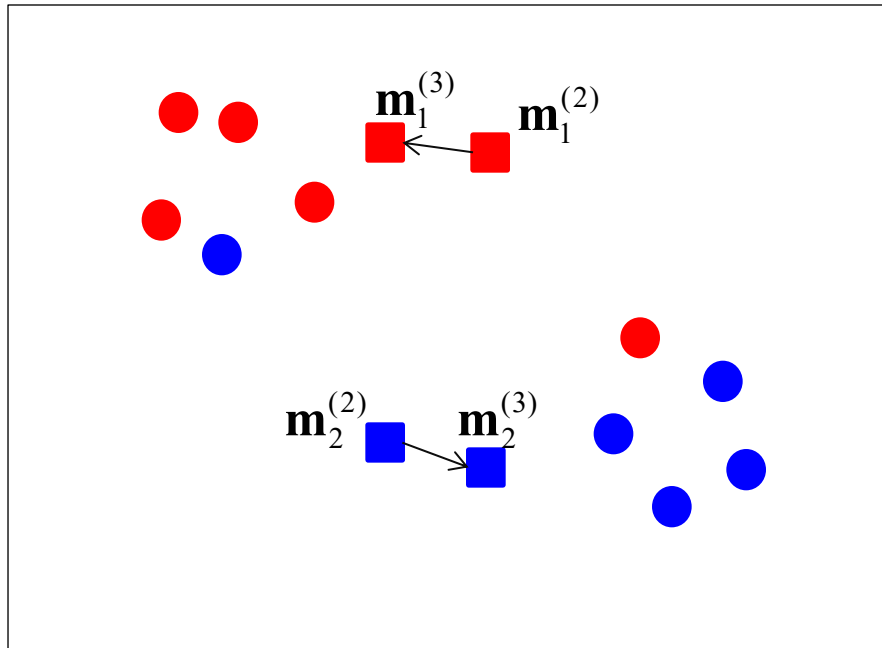
Assign all the other points to the centers in the same way as $\mathbf{x}_p$

$$S_1^{(2)} = \{\mathbf{x}_p : \|\mathbf{x}_p - \mathbf{m}_1^{(2)}\|^2 \leq \|\mathbf{x}_p - \mathbf{m}_2^{(2)}\|^2\}$$

$$S_2^{(2)} = \{\mathbf{x}_p : \|\mathbf{x}_p - \mathbf{m}_2^{(2)}\|^2 \leq \|\mathbf{x}_p - \mathbf{m}_1^{(2)}\|^2\}$$

K means: details

Second iteration



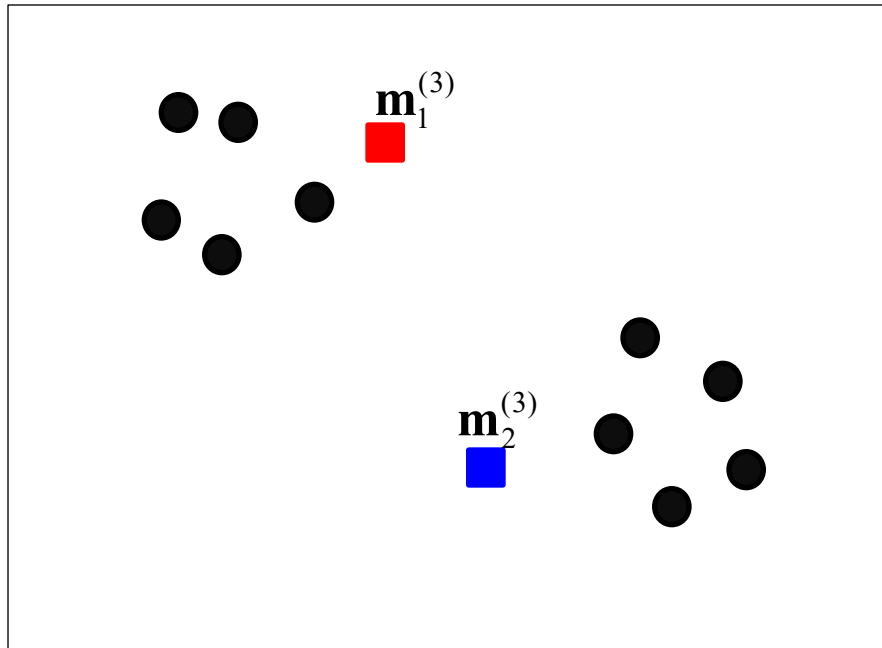
Update new centers by averaging the data points in each set:

$$\mathbf{m}_1^{(3)} = \sum_{\mathbf{x}_p \in S_1^{(2)}} \mathbf{x}_p / |S_1^{(2)}|$$

$$\mathbf{m}_2^{(3)} = \sum_{\mathbf{x}_p \in S_2^{(2)}} \mathbf{x}_p / |S_2^{(2)}|$$

K means: details

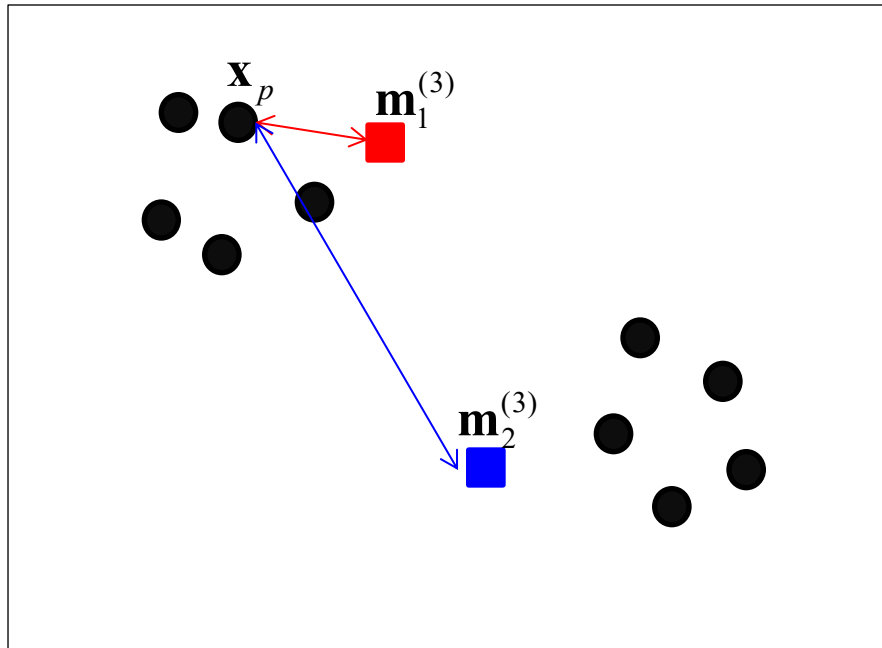
Second iteration



The result after second iteration

K means: details

Third iteration



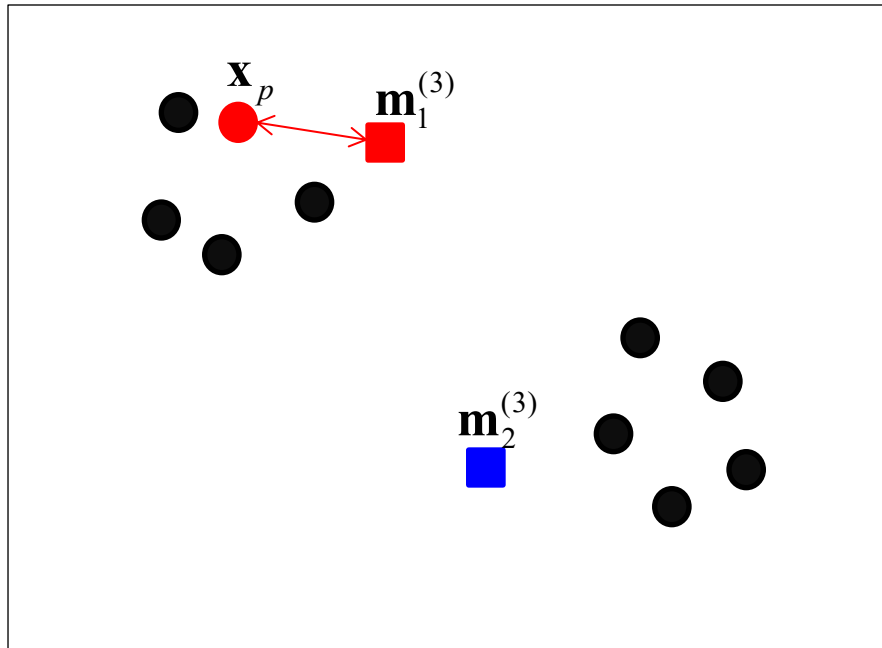
Calculate the Euclidean distance between each point and the centers

$$d^2(\mathbf{x}_p, \mathbf{m}_1^{(3)}) = \|\mathbf{x}_p - \mathbf{m}_1^{(3)}\|^2 = \sum_i (x_{pi} - m_{1i}^{(3)})^2$$

$$d^2(\mathbf{x}_p, \mathbf{m}_2^{(3)}) = \|\mathbf{x}_p - \mathbf{m}_2^{(3)}\|^2 = \sum_i (x_{pi} - m_{2i}^{(3)})^2$$

K means: details

Third iteration



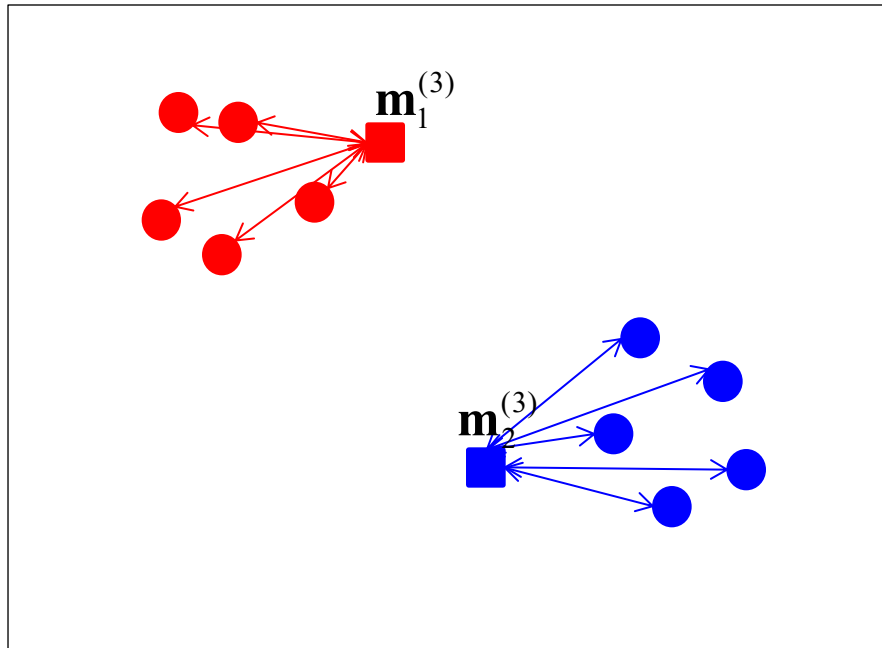
Assign the data point to the center with smallest distance:

$$d^2(\mathbf{x}_p, \mathbf{m}_1^{(3)}) < d^2(\mathbf{x}_p, \mathbf{m}_2^{(3)})$$

Then $\mathbf{x}_p$ is assigned to $\mathbf{m}_1^{(3)}$

K means: details

Third iteration



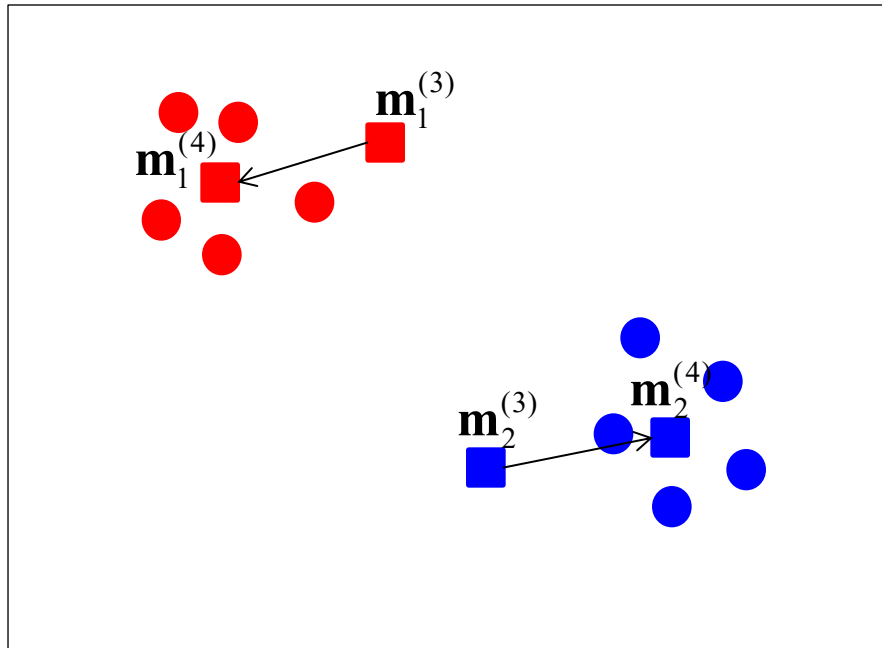
Assign all the other points to the centers in the same way as $\mathbf{x}_p$

$$S_1^{(3)} = \{\mathbf{x}_p : \|\mathbf{x}_p - \mathbf{m}_1^{(3)}\|^2 \leq \|\mathbf{x}_p - \mathbf{m}_2^{(3)}\|^2\}$$

$$S_2^{(3)} = \{\mathbf{x}_p : \|\mathbf{x}_p - \mathbf{m}_2^{(3)}\|^2 \leq \|\mathbf{x}_p - \mathbf{m}_1^{(3)}\|^2\}$$

K means: details

Third iteration



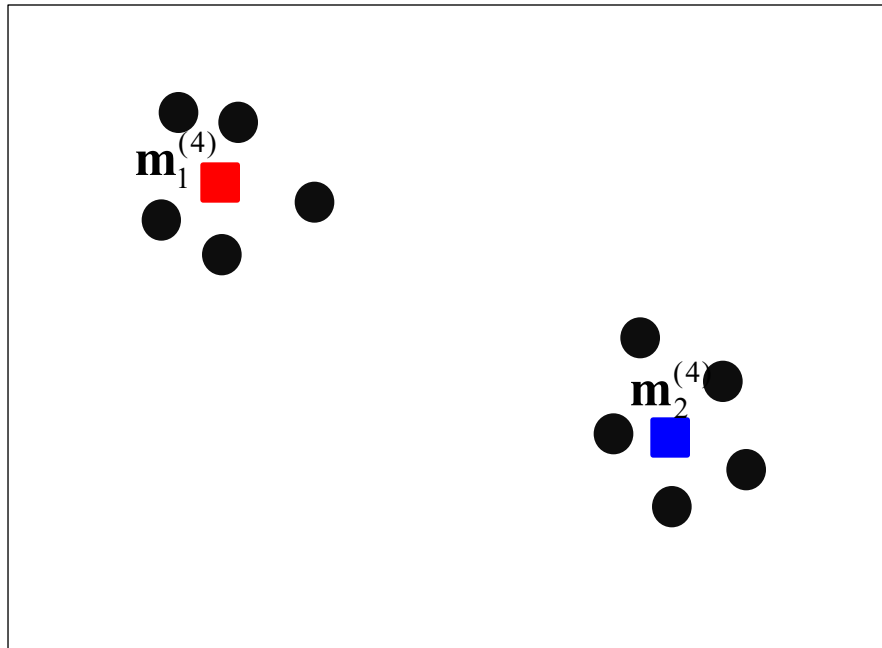
Update new centers by averaging the data points in each set:

$$\mathbf{m}_1^{(4)} = \sum_{\mathbf{x}_p \in S_1^{(3)}} \mathbf{x}_p / |S_1^{(3)}|$$

$$\mathbf{m}_2^{(4)} = \sum_{\mathbf{x}_p \in S_2^{(3)}} \mathbf{x}_p / |S_2^{(3)}|$$

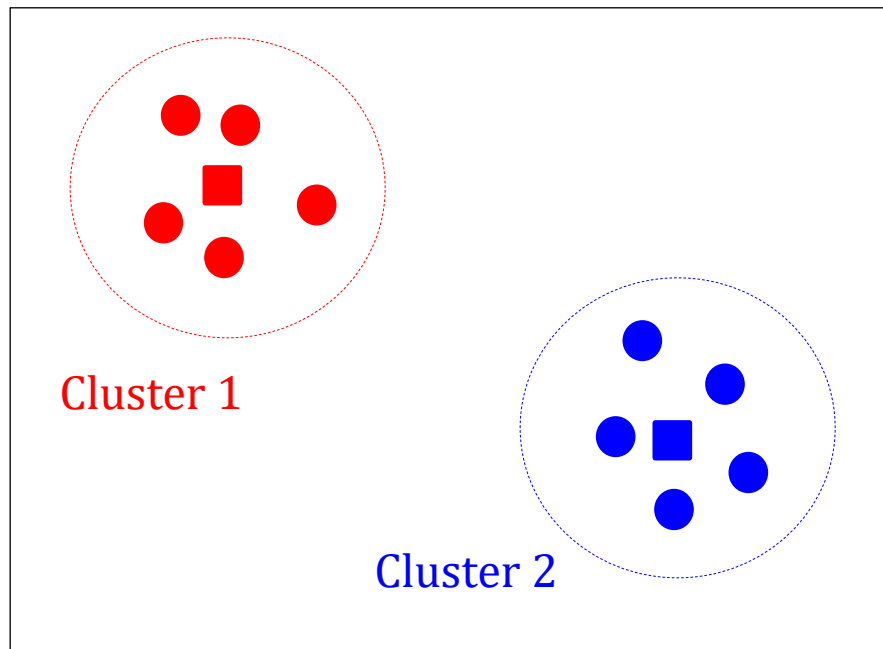
K means: details

Forth iteration



Converge !

K means: details



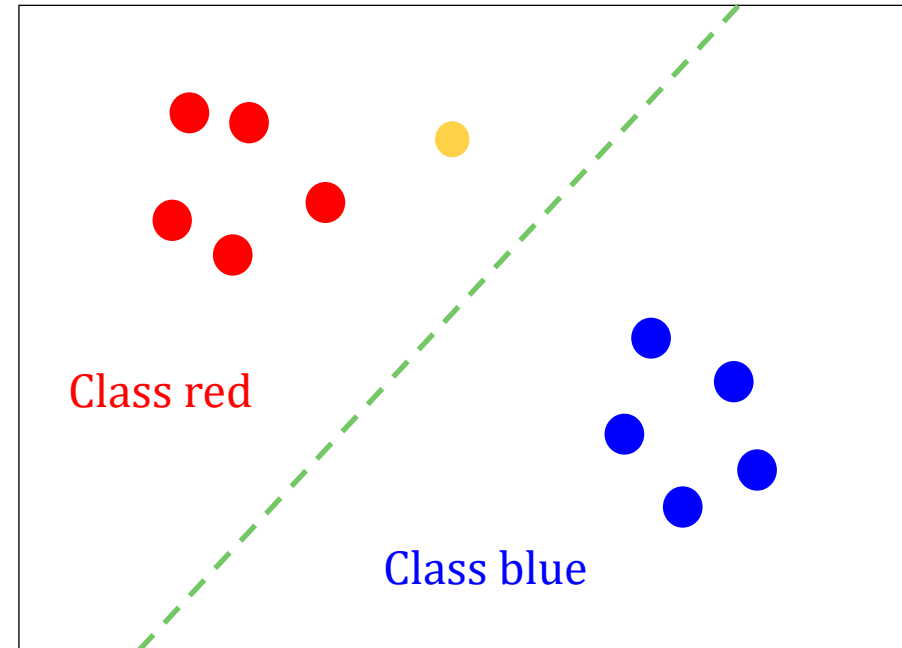
Clustering result

Agenda

- Machine learning
- Clustering
 - k means clustering
- Classification

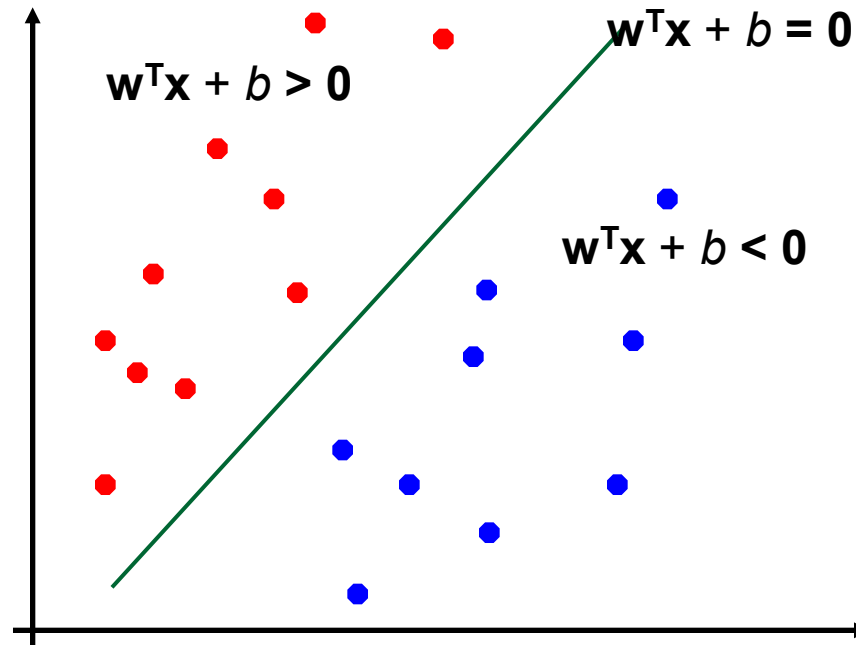
Classification

- Classification
 - Assign the data to the predefined label
- Training data
 - A set of data with predefined labels
- Test data
 - A set of data, which is used to validate the classification performance
 - The labels of the data are defined as ground truth
 - The labels from the computational model are defined as the classification results



Linear Classification

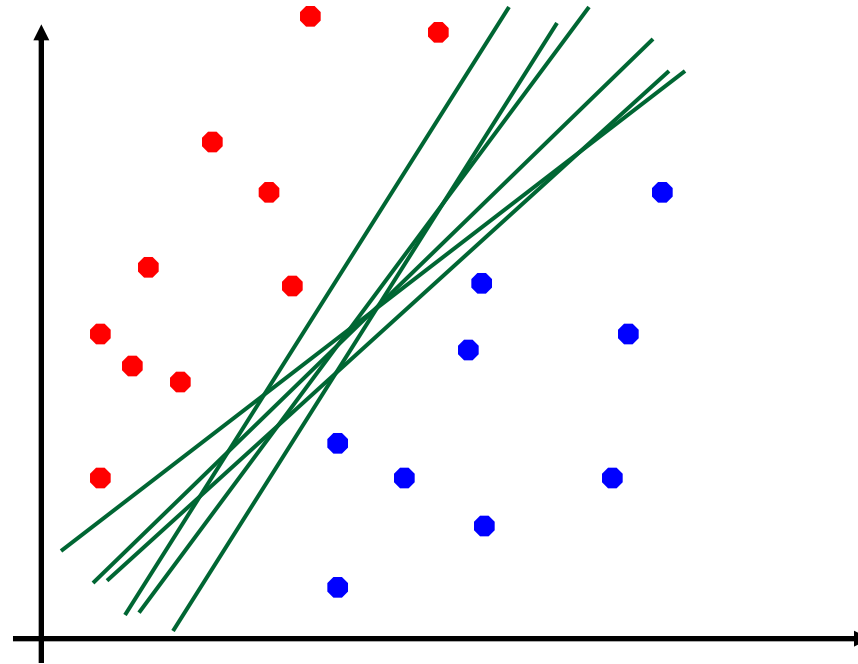
- Linear classification can be viewed as the task of finding the linear separator that can separate different classes in the feature space:



$$f(\mathbf{x}) = \text{sign}(w^T \mathbf{x} + b)$$

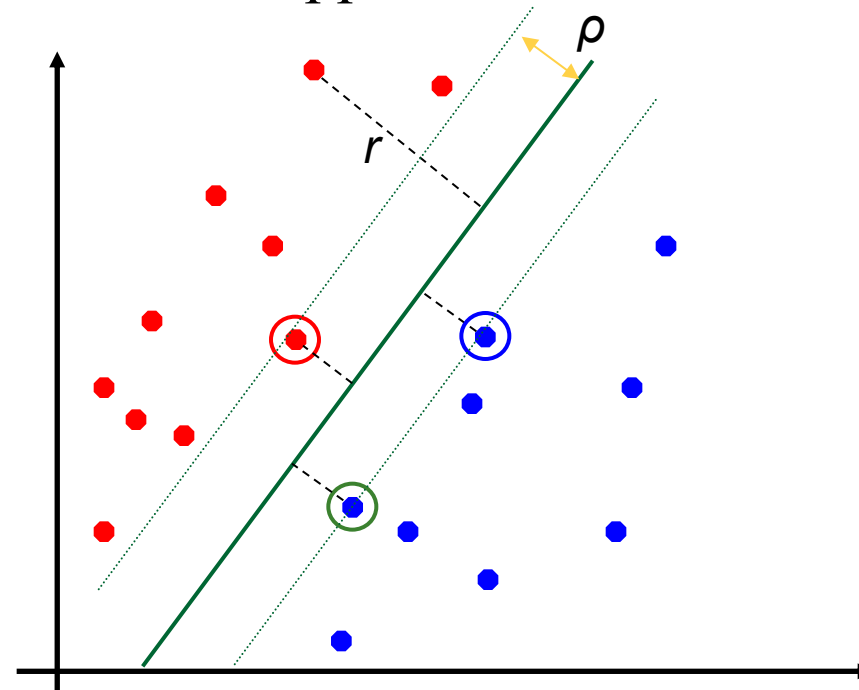
Multiple Solutions

- Which of the linear separators is the best?



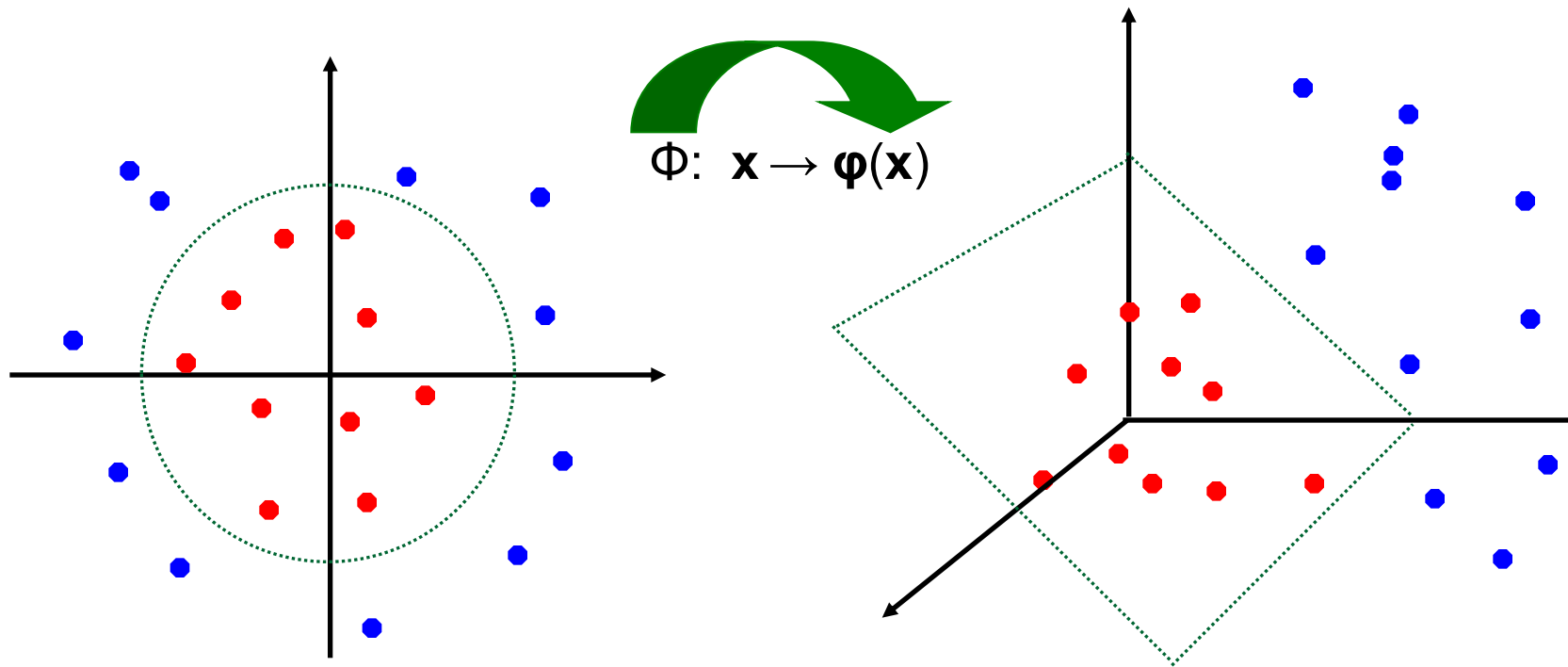
Classification Margin

- Distance from example $\mathbf{x}_i$ to the separator is $r = |(\mathbf{w}^T \mathbf{x}_i + b)| / \|\mathbf{w}\|$
- Data points closest to the hyperplane are *support vectors*.
- *Margin* ρ of the separator is the distance between support vectors and the classification boundary
- Example
 - Given two data points
 - (5, 1) belonging to class red
 - (-1, -1) belonging to class blue
 - Calculate w_1 and w_0
 - $Y = -3X + 6$



Nonlinear SVM

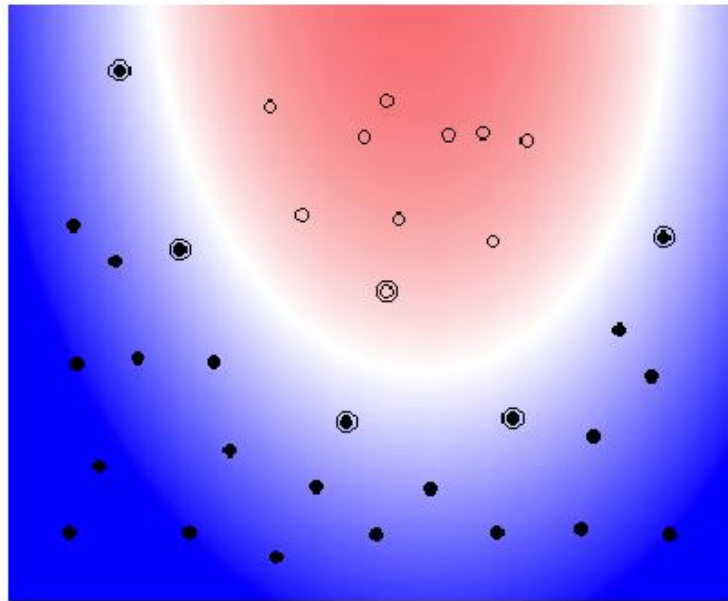
- General idea
 - The original feature space can always be mapped to some higher-dimensional feature space where the training set is separable



Examples of SVM with Different Kinds of Kernels

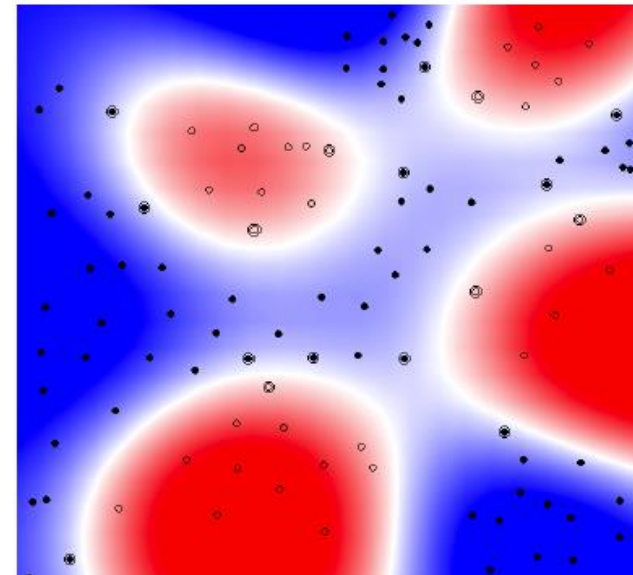
- Data points and classification boundary mapped to the feature space of the original data

Polynomial Kernel



$$K(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) = (\mathbf{x}^{(i)\top} \mathbf{x}^{(j)} + 1)^d$$

Radial-basis Function Kernel



$$K(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) = \exp\left(-\frac{\|\mathbf{x}^{(i)} - \mathbf{x}^{(j)}\|^2}{2\sigma^2}\right)$$